

The problem: Deployment

complexity barrier

From localhost to Agent Engine

In part 1, you set up your Google Cloud environment.
Now you'll deploy your first agent to production.

Where you are now:

- ✓ Google Cloud account ready
- ✓ APIs enabled (Vertex AI, Cloud Run)
- ✓ gcloud CLI authenticated
- ✓ Vertex AI SDK installed

Where you're going:

None

Local development: `adk web → http://localhost:8000`



Agent Engine: `adk deploy agent-engine → https://agent-xyz.googleapis.com`

This part: Transform your localhost agent into a production service with one command.

The problem:

Deployment complexity barrier

Traditional cloud deployment is complex:

- Write Dockerfiles
- Configure Kubernetes
- Set up databases for session state
- Manage infrastructure
- Handle scaling and load balancing

Students are agent developers, not DevOps engineers

Concrete example:

Python

```
# Course 8 multi-agent workflow (works locally)
from google.adk.agents import SequentialAgent, ParallelAgent, LoopAgent

workflow = SequentialAgent(
    name='content_qa_system',
    sub_agents=[intake, fact_check_workflow, quality_loop]
)

# Local testing: adk web ✅
# Production deployment: ??? ❌
# Need: Docker, Kubernetes, databases, infrastructure...
```

The root problem

Students need one-command deployment without infrastructure complexity.